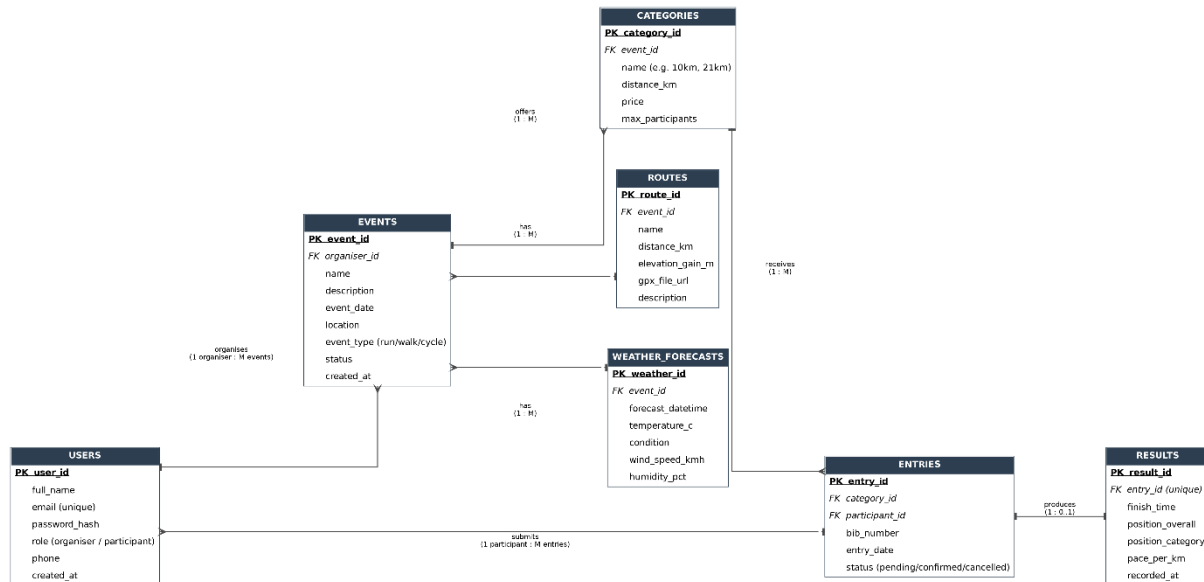


STUDENT NO.:ST10470734

NAME:PFUNZO MULAUDZI

SECTION A



SECTION B

Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Responses
POST	/api/auth/register	Creates a new user account as either an organiser or participant	NONE	{ fullName, email, password, role, phone }	201 Created user objects; 409 Conflict email already registered
POST	/api/auth/login	Authenticates a user and issues an access token.	NONE	{ email, password }	200 OK { token, user }; 401 Unauthorized invalid credentials

USER PROFILE

GET	/api/users/me	Returns the logged-in user's own profile.	Any	NONE	200 OK user profile object
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PUT	/api/users/me	Updates the logged-in user's profile details.	Any	{ fullName, phone }	200 OK updated profile; 400 Bad Request invalid fields
GET	/api/users/me/history	Returns the participant's personal performance history, including past entries and results.	Participant	NONE	200 OK array of past entries with linked results

EVENTS

GET	/api/events	Lists upcoming events and supports filtering such as type and location.	NONE	NONE	200 OK array of event summaries
GET	/api/events/{id}	Returns full details for one event, including categories and routes.	NONE	NONE	200 OK event object; 404 Not Found event doesn't exist
POST	/api/events	Creates a new event owned by the logged-in organiser.	OWNER	{ name, description, eventDate, location, eventType }	201 Created new event; 400 Bad Request validation failed
PUT	/api/events/{id}	Updates an existing event's details.	OWNER	{ name, description, eventDate, location, eventType, status }	200 OK updated event; 403 Forbidden not owner; 404 Not Found

DELETE	/api/events/{id}	Cancels or removes an event.	OWNER	NONE	200 OK confirmation; 403 Forbidden; 404 Not Found
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CATEGORIES

GET	/api/events/{id}/categories	/api/categories/{id}	NONE	NONE	200 OK array of categories
POST	/api/events/{id}/categories	Adds a new category to an event.	OWNER	{ name, distanceKm, price, maxParticipants }	201 Created new category; 403 Forbidden; 404 Not Found
PUT	/api/categories/{id}	Updates a category's details.	OWNER	{ name, distanceKm, price, maxParticipants }	200 OK updated categories; 403 Forbidden; 404 Not Found
DELETE	/api/categories/{id}	Removes a category.	OWNER	NONE	200 OK confirmation; 403 Forbidden; 404 Not Found

ENTRIES

POST	/api/categories/{id}/entries	Enrols the logged-in participant into a category. The participant is identified from the authentication token.	Participant	NONE	201 Created entry with generated bib number and "pending" status; 404 Not Found category doesn't exist; 409 Conflict category full
------	------------------------------	--	-------------	------	--

					or already entered
GET	/api/users/me/entries	Lists the logged-in participant's own past and upcoming entries.	Participant	NONE	200 OK array of entries
GET	/api/events/{id}/entries	Lists all entries for an event as the organiser's roster/start list.	Owner	NONE	200 OK entries with participant details; 403 Forbidden
DELETE	/api/entries/{id}	Cancels an entry.	Participant or Owner	NONE	200 OK entries with participant details; 403 Forbidden

RESULTS

POST	/api/entries/{id}/results	Captures a finish-line result for one entry.	Owner	{ finishTime, positionOverall, positionCategory }	201 Created result object; 403 Forbidden; 404 Not Found; 409 Conflict result already recorded
GET	/api/entries/{id}/results	Returns the leaderboard for a category, sorted by finish position.	None	NONE	200 OK ordered array of results
GET	/api/users/me/results	Returns the logged-in participant's own result history.	Participant	NONE	200 OK array of results linked to participant's entries

Routes and Weather

GET	/api/events/{id}/routes	Lists routes for an event to help participants plan for race day.	NONE	NONE	200 OK array of routes
-----	-------------------------	---	------	------	------------------------

POST	/api/events/{id}/routes	Adds a route to an event.	OWNER	{ name, distanceKm, elevationGainM, gpxFileUrl, description }	201 Created new route; 403 Forbidden; 404 Not Found
GET	/api/events/{id}/weather	Returns the latest weather forecast for an event's date and location.	NONE	NONE	200 OK forecast object; 404 Not Found no forecast recorded

SECTION 3

```
CREATE DATABASE RaceDayDB;
```

```
USE RaceDayDB;
```

```
CREATE TABLE Users (
```

```
    user_id INT IDENTITY PRIMARY KEY,
```

```
    full_name VARCHAR(100) NOT NULL,
```

```
    email VARCHAR(255) NOT NULL,
```

```
    password_hash VARCHAR(255) NOT NULL,
```

```
    role VARCHAR(20) NOT NULL DEFAULT 'participant',
```

```
    phone VARCHAR(20) NULL,
```

```
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
```

```
    CONSTRAINT UQ_Users_Email UNIQUE (email),
```

```
    CONSTRAINT CK_Users_Role CHECK (role IN
```

```
    ('organiser', 'participant')));
```

```
CREATE TABLE Events (
```

```
    event_id INT IDENTITY PRIMARY KEY,
```

```
    organiser_id INT NOT NULL,
```

```
    name VARCHAR(150) NOT NULL,
```

```

description TEXT NULL,

event_date DATE NOT NULL,

location VARCHAR(150) NOT NULL,

event_type VARCHAR(20) NOT NULL,

status VARCHAR(20) NOT NULL DEFAULT 'upcoming',

created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT FK_Events_Organiser FOREIGN KEY (organiser_id) REFERENCES
Users (user_id),

CONSTRAINT CK_Events_Type CHECK (event_type IN ('running', 'walking',
'cycling')),

CONSTRAINT CK_Events_Status CHECK (status IN ('upcoming', 'completed',
'cancelled')) );

```

```

CREATE TABLE Categories (

    category_id INT IDENTITY PRIMARY KEY,

    event_id INT NOT NULL, name VARCHAR(50) NOT NULL,

    distance_km DECIMAL(5,2) NOT NULL,

    price DECIMAL(8,2) NOT NULL DEFAULT 0,

    max_participants INT NOT NULL DEFAULT 0,

    CONSTRAINT FK_Categories_Event FOREIGN KEY (event_id) REFERENCES
Events (event_id),

    CONSTRAINT UQ_Categories_EventName UNIQUE (event_id, name),

    CONSTRAINT CK_Categories_Distance CHECK (distance_km > 0),

    CONSTRAINT CK_Categories_MaxParticipants CHECK (max_participants >= 0) );

```

```

CREATE TABLE Entries (

    entry_id INT IDENTITY PRIMARY KEY,

    category_id INT NOT NULL,

    participant_id INT NOT NULL,

```

```

        bib_number VARCHAR(10) NULL,

        entry_date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

        status VARCHAR(20) NOT NULL DEFAULT 'pending',

        CONSTRAINT FK_Entries_Category FOREIGN KEY (category_id) REFERENCES
Categories (category_id),

        CONSTRAINT FK_Entries_Participant FOREIGN KEY (participant_id) REFERENCES
Users (user_id),

        CONSTRAINT UQ_Entries_CategoryParticipant UNIQUE (category_id,
participant_id),

        CONSTRAINT CK_Entries_Status CHECK (status IN ('pending', 'confirmed',
'cancelled')) );

```

```

CREATE TABLE Results (

        result_id INT IDENTITY PRIMARY KEY,

        entry_id INT NOT NULL,

        finish_time TIME NULL,

        position_overall INT NULL,

        position_category INT NULL,

        pace_per_km TIME NULL,

        recorded_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

        CONSTRAINT FK_Results_Entry FOREIGN KEY (entry_id) REFERENCES Entries
(entry_id),

        CONSTRAINT UQ_Results_Entry UNIQUE (entry_id) );

```

```

CREATE TABLE Weather_Forecasts (

        weather_id INT IDENTITY PRIMARY KEY,

        event_id INT NOT NULL,

        forecast_datetime DATETIME NOT NULL,

        temperature_c DECIMAL(4,1) NULL,

```

```
        condition_text VARCHAR(50) NULL,  
        wind_speed_kmh DECIMAL(5,1) NULL,  
        humidity_pct INT NULL,  
        CONSTRAINT FK_Weather_Event FOREIGN KEY (event_id) REFERENCES Events  
(event_id),  
        CONSTRAINT CK_Weather_Humidity CHECK (humidity_pct IS NULL OR  
humidity_pct BETWEEN 0 AND 100) );
```

```
INSERT INTO Users (  
    full_name, email, password_hash, role, phone)  
VALUES ('Thandiwe Nkosi', 'thandiwe.nkosi@raceday.co.za', 'hashed_pw_001',  
'organiser', '0721234567'),  
('Pieter van Wyk', 'pieter.vanwyk@raceday.co.za', 'hashed_pw_002', 'organiser',  
'0839876543'),  
('Lindiwe Dube', 'lindiwe.dube@example.com', 'hashed_pw_003', 'participant',  
'0731122334'),  
('Sipho Mahlangu', 'sipho.mahlangu@example.com', 'hashed_pw_004',  
'participant', '0665544332');
```

```
INSERT INTO Events (  
    organiser_id, name, description, event_date, location, event_type, status)  
VALUES (1, 'Comrades Marathon', 'Iconic ultramarathon between  
Pietermaritzburg and Durban.', '2027-06-13', 'Pietermaritzburg, KZN', 'running',  
'upcoming'),  
(1, 'Soweto Marathon', 'Community road race through the streets of Soweto.',  
'2027-02-14', 'Soweto, Gauteng', 'running', 'upcoming'),  
(2, 'Cape Town Cycle Tour', 'One of the worlds largest timed cycling events.',  
'2027-03-08', 'Cape Town, Western Cape', 'cycling', 'upcoming');
```

```
INSERT INTO Categories (  
    event_id, name, distance_km, price, max_participants)
```



```
VALUES (1, '90km Ultra', 90.00, 950.00, 20000),  
(1, 'Novice 56km', 56.00, 950.00, 5000),  
(2, '10km', 10.00, 180.00, 3000),  
(2, '21km', 21.10, 250.00, 3000),  
(3, '109km Full Tour', 109.00, 650.00, 35000),  
(3, '56km Half Tour', 56.00, 450.00, 10000);
```

```
INSERT INTO Weather_Forecasts (  
    event_id, forecast_datetime, temperature_c, condition_text,  
    wind_speed_kmh, humidity_pct)  
VALUES (1, '2027-06-13 05:30:00', 9.5, 'Clear', 10.0, 55),  
(3, '2027-03-08 06:00:00', 18.0, 'Partly cloudy', 22.5, 68);
```

```
INSERT INTO Entries (  
    category_id, participant_id, bib_number, status)  
VALUES (1, 3, '10045', 'confirmed'); -- Lindiwe entered Comrades 90km Ultra  
(3, 3, '20871', 'confirmed'); -- Lindiwe also entered Soweto 10km (4, 4, '20872',  
'pending'), -- Sipho entered Soweto 21km (5, 4, '30456', 'confirmed'); -- Sipho entered  
Cape Town Cycle Tour 109km
```

```
INSERT INTO Results (  
    entry_id, finish_time, position_overall, position_category, pace_per_km)  
VALUES (2, '00:48:32', 154, 12, '00:04:51'); -- Lindiwe's Soweto 10km result (4,  
'02:52:10', 890, 340, '00:04:53'); -- Sipho's Cape Town Cycle Tour
```

```
SELECT * FROM Users;
```

```
SELECT * FROM Events;
```

```
SELECT * FROM Categories;
```

```
SELECT * FROM Entries;
```

```
SELECT * FROM Results;
```

```
SELECT * FROM Weather_Forecasts;
```